

```
1 --s21_b
2
3 LIBRARY ieee;
4 USE ieee.std_logic_1164.ALL;
5 USE ieee.numeric_std.ALL;
6
7 ENTITY s21_b IS
8     PORT (
9         clk : IN STD_LOGIC;
10        u : IN STD_LOGIC;
11        y : OUT STD_LOGIC
12    );
13END ENTITY;
14
15ARCHITECTURE behavioral OF s21_b IS
16    SIGNAL q : STD_LOGIC_VECTOR(4 DOWNTO 0) := "00000";
17BEGIN
18
19    -- sequential process
20    seq_p : PROCESS (clk)
21    BEGIN
22        IF rising_edge(clk) THEN
23            -- making the shift register
24            q <= u & q(4 DOWNTO 1);
25            IF q = "10100" THEN
26                y <= '1';
27            ELSE
28                y <= '0';
29            END IF;
30        END IF;
31    END PROCESS;
32
33END ARCHITECTURE behavioral;
```